

Course Project Guide

Generative Modeling: Diffusion, Flow Matching, and Stochastic Interpolants

Project format

The course project is a small reproduction of a published research result, with one extension. It will be performed in teams of two to three students. The teams will choose a project from a list of approved track–paper bundles (see below). A bundle consists of the main paper or paper pair, the default dataset, and the range of acceptable extensions.

Concretely: each team should reproduce one baseline result or research claim at modest scale, make one change (theoretical / architectural / data), and evaluate that change carefully. The goal is not to invent new stuff or read many papers; rather, you should read a real paper closely, implement a clean baseline, learn how to make and defend a small empirical comparison, and then present clearly to your colleagues.

The final submission will consist of a short report, a runnable repository, and a short presentation.

How bundles work

The class project menu is organized into five tracks, with two bundles per track; this means there are ten bundles. Each bundle below names the paper or paper pair that anchors the project. Each team will submit four ranked bundle preferences (which must belong to different tracks), together with a short note about the team member background, available compute, and anticipated compute requirements. We will then assign one bundle to each team so as to maximize happiness. Each bundle will be assigned to at most one team.

This system is meant to give you choice while keeping the overall project manageable and presentations diverse.

Scope

Keep the project close to the assigned bundle. Use one dataset, ask one main question, and vary one extension axis. If the bundle is a direct comparison bundle, it may involve two model families because that comparison is the point of the project. Otherwise, use one model (model family).

The final report should be based on a small set of experiments. In most cases that means a baseline, the chosen extension, one control or alternate setting, and one failure or sanity case. Do not add extra datasets, extra papers, or unrelated ablations. Again, the point is that you get familiar with the practice of generative diffusion models.

Project menu

Unless a bundle description says otherwise, the expected scale is either toy two-dimensional data or one of the shared image datasets prepared for the course. For most image bundles, the default dataset is CIFAR-10. CelebA-64 is the main higher-quality visual dataset and is especially suitable for unconditional image-generation bundles. MNIST and Fashion-MNIST remain available as fallback datasets if a team needs a lighter configuration after discussion with the course staff.

Code	Bundle	Default dataset	Scale
A1	DDPM and Flow Matching	CIFAR-10	image
A2	DDPM and Rectified Flow	CelebA-64	image
B1	DDPM and DDIM	CIFAR-10	image
B2	DDPM and DPM-Solver	CelebA-64	image
C1	Conditional diffusion with classifier guidance	CIFAR-10	image
C2	Conditional diffusion with classifier-free guidance	CIFAR-10	image
D1	Denoising score matching and annealed Langevin	2D toy data	toy
D2	Annealed Langevin and the SDE/ODE view	2D toy data	toy
E1	Stochastic interpolants: deterministic vs. stochastic interpolation	2D toy data	toy
E2	Stochastic interpolants and Flow Matching on toy data	2D toy data	toy

CIFAR-10 is the standard benchmark for most image projects. CelebA-64 is the nicer visual dataset suitable mainly for unconditional bundles. (Though feel free to find other datasets if you're annoyed with celebrity faces; we understand.) Guidance bundles should probably stick with CIFAR-10 because the class labels make conditional generation and evaluation simpler.

Track A: comparison projects

A1. DDPM and Flow Matching. Papers: Ho et al., *Denoising Diffusion Probabilistic Models*, and Lipman et al., *Flow Matching for Generative Modeling*. This bundle compares a small diffusion baseline to a matched Flow Matching model on CIFAR-10. The main task is to implement both methods at comparable scale and compare them fairly. Suitable extensions include a fixed-NFE comparison, a fixed wall-clock comparison, or a small stability comparison across seeds.

A2. DDPM and Rectified Flow. Papers: Ho et al., *Denoising Diffusion Probabilistic Models*, and Liu et al., *Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow*. This bundle compares a small DDPM to a small rectified-flow model on CelebA-64. The work should stay focused on comparing two methods. Suitable extensions include solver steps, wall-clock cost, or stability.

Track B: faster diffusion sampling

B1. DDPM and DDIM. Papers: Ho et al., *Denoising Diffusion Probabilistic Models*, and Song et al., *Denoising Diffusion Implicit Models*. This bundle begins with a small DDPM on CIFAR-10 and compares the standard sampler to DDIM. Suitable extensions include a step-count sweep, a small schedule study, or a comparison between deterministic and stochastic sampling.

B2. DDPM and DPM-Solver. Papers: Ho et al., *Denoising Diffusion Probabilistic Models*, and Lu et al., *DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling*. This bundle also begins with a small DDPM on CelebA-64 and compares the standard sampler to DPM-Solver. Suitable extensions include solver order, step count, or schedule interaction.

Track C: guidance and conditioning

C1. Conditional diffusion with classifier guidance. Papers: Ho et al., *Denoising Diffusion Probabilistic Models*, and Dhariwal & Nichol, *Diffusion Models Beat GANs on Image Synthesis*. This bundle studies class-conditional generation on CIFAR-10 with an auxiliary classifier. Suitable

extensions include a guidance-scale sweep, a simple guidance schedule, or a study of reduced-data classifier training.

C2. Conditional diffusion with classifier-free guidance. Papers: Ho et al., *Denoising Diffusion Probabilistic Models*, and Ho & Salimans, *Classifier-Free Diffusion Guidance*. This bundle studies class-conditional generation on CIFAR-10 with classifier-free guidance. Suitable extensions are usually a guidance-scale sweep or a simple guidance schedule.

Track D: score-based bridge

This and the next track use toy datasets but you should feel free to also use real image datasets if they help you prove or illustrate your point, or just for fun. This will be viewed favorably, but only after the core toy experiments are in place.

D1. Denoising score matching and annealed Langevin. Papers: Vincent, *A Connection Between Score Matching and Denoising Autoencoders*, and Song & Ermon, *Generative Modeling by Estimating Gradients of the Data Distribution*. This bundle works on toy two-dimensional data. The baseline is a score network trained with denoising score matching and sampled with annealed Langevin dynamics. Suitable extensions include the number of noise levels, the number of Langevin steps, or a comparison between score quality and sample quality.

D2. Annealed Langevin and the SDE/ODE view. Papers: Song & Ermon, *Generative Modeling by Estimating Gradients of the Data Distribution*, and Song et al., *Score-Based Generative Modeling through Stochastic Differential Equations*. This bundle compares the older annealed-Langevin view to the reverse-SDE or probability-flow-ODE view.

Track E: stochastic interpolants

E1. Stochastic interpolants: deterministic vs. stochastic interpolation. Paper: Albergo, Boffi, and Vanden-Eijnden, *Stochastic Interpolants: A Unifying Framework for Flows and Diffusions*. This bundle stays within the stochastic-interpolants framework on toy data. The baseline is one small result or figure from the paper, and the extension studies one design choice inside SI itself, such as deterministic versus stochastic interpolation or a change in interpolant choice.

E2. Stochastic interpolants and Flow Matching on toy data. Papers: Albergo, Boffi, and Vanden-Eijnden, *Stochastic Interpolants: A Unifying Framework for Flows and Diffusions*, and Lipman et al., *Flow Matching for Generative Modeling*. This bundle compares the two viewpoints on the same toy problem. Suitable extensions include interpolant choice, solver cost, or a small change in the stochastic component. This bundle is conceptually demanding and should remain small.

Calendar

The aim is to get the code running early and to fix scope by the middle of the project.

Week	Stage	What should be done by the end of the week
1	Choice, proposal, and smoke test	Team formed, bundle preferences submitted, bundle assigned, proposal drafted, repository created, environment runs, data pipeline works, and at least one end-to-end run has completed.
2	Baseline work	Baseline implementation in place and first results compared to the paper.
3	First ablation and scope freeze	One baseline result and one meaningful comparison completed; final experiment matrix fixed.
4	Main experiments	Remaining core runs completed and main figures taking shape.
5	Writing and cleanup	Report drafted, figures cleaned up, repository organized, and reproduction path checked.
6	Submission and talk	Final report, repository, and presentation submitted.

Deliverables

The proposal should be one page. Keep it close to the spirit of the assigned bundle, without extending the project. The proposal should show that the team understands the baseline, the proposed extension, the evaluation plan, and the compute budget. Try to have a smoke test completed soon after: verify that the repository runs end-to-end on small data.

The mid-project checkpoint comes at the scope-freeze point. The team should have an approximate baseline reproduction, one first ablation, and a short explanation of where its results differ from the paper.

The final submission consists of a short report, a reproducible repository, and a presentation. The report should be four to six pages of main text. It should describe the baseline, the extension, the evaluation setup, the main results, and the main limitations. The repository should include setup instructions, dependencies, and a clear command to reproduce the main figure and/or table. The presentation should be concise; exact length will be communicated later, but for the moment think 15–20 minutes.